

RUDRA SHUKLA

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TECHNICAL SKILLS

Languages: Java, Python, C/C++

Hardware: ESP32, Arduinos, Circuit Boards, Raspberry Pi

Tools: AutoCAD, SolidWorks, PLC, Soldering, 3D Printing, Google Cloud Services, Linux, Canva, Microsoft Suites

PROJECTS

Electronics Keypad Vault

July 2025

Arduino, C/C++, 3D Designing, Soldering

Ajax, ON

- Engineered a secure electronic vault system using Arduino and C/C++, integrating a custom keypad interface with password authentication and fail-safe logic for enhanced security.
- Designed and fabricated a custom enclosure using 3D modeling software, optimizing for component layout, durability, and aesthetic appeal before 3D printing.

Car Rental System

June 2025

Java, OOP

Ajax, ON

- Developed a comprehensive car rental application using Java, applying object-oriented principles such as inheritance and encapsulation to manage vehicles, customers, and rental workflows.
- Optimized code structure through reusable classes and interfaces, enhancing scalability and maintainability while demonstrating proficiency in clean architecture and software design patterns.

LED Sign Board

January 2025

ESP32, C/C++, WLED, Soldering

Ajax, ON

- Designed and built a custom 28x8 LED Sign Board controlled via IPv4 address using WLED firmware and C/C++ programming on an ESP32 microcontroller.
- Utilized CNC machining to precisely cut the signboard, ensuring optimal fit and aesthetics while facilitating efficient assembly and executed soldering of connections.

WORK/VOLUNTEER EXPERIENCE

Robotics Instructor

January 2023 – August 2025

OBOTZ - IQ Brainers Academy

Ajax, ON

- Formulated and implemented engaging robotics curricula for **20+** students of varying skill levels, fostering hands-on learning and teamwork in building and programming robots in Arduino.
- Facilitated workshops and competitions, mentoring students in problem-solving, critical thinking, and coding, resulting in increased participation **50%** and success in robotics events.
- Collaborated with faculty and industry professionals to integrate robotics concepts such as obstacle avoider, line follower, into existing STEM programs, enhancing student understanding of real-world applications.

Warrant Officer First Class (WO1) & Senior Instructor

September 2023 – June 2025

856 Royal Canadian Air Cadets

Pickering, ON

- Led leadership training, fostering discipline and teamwork while enhancing **200+** cadets' decision-making skills.
- Completed advanced aviation training, including flight theory, navigation, and aircraft systems demonstrating technical aptitude and discipline in high-responsibility environments.
- Managed training night schedules and supervised cadets, ensuring adherence to uniform standards while addressing infractions to maintain professionalism.

Mechanical and Design Lead - FIRST Tech Challenge

September 2021 – June 2025

J. Clarke Richardson Collegiate

Ajax, ON

- Constructed robot components using CAD tools, integrating structural analysis to optimize performance.
- Led mechanical subsystem development, including drivetrain, arm mechanisms, and intake systems, applying principles of torque, gear ratios, and material selection to enhance reliability and efficiency.
- Collaborated with software and strategy teams to align mechanical design with autonomous and driver-controlled functionality, resulting in a **25%** increase in scoring consistency across regional matches.

EDUCATION

University of Waterloo

Waterloo, ON

Candidate for B.A.Sc in Mechatronics Engineering

September 2025 – Present

- Relevant Coursework:** Digital Computation, Linear Algebra, Calculus 1, Engineering Graphics and Design
- Design Team:** University of Waterloo Alternative Fuels Team (UWAFI)